

Reinforcement Learning

Assignment 1

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1 Assignment 1: Reinforcement Learning

1.1 Question 1

Q1: “Prove that the policy iteration algorithm converges to the optimal policy for a finite MDP with a finite action space for each state.”

First of all, its good to define each key word:

Optimal Policy: An optimal policy is a strategy or decision-making rule that maximizes the expected cumulative reward for an agent in a given environment. An optimal policy is one that leads to the highest possible long-term reward for the agent, considering the dynamics of the environment and the available actions.

Finite MDP: A Markov Decision Process (MDP) is used to model the decision-making problems where an agent interacts with an environment over discrete time steps. The keyword "finite" only means that the MDP consists of a limited set of states and actions with respective transition function that defines the probabilities of moving from one state to another given an action, and a reward function that assigns a numerical value to each state-action pair.

Finite Action Space: A finite action space refers to a situation in which an agent has a limited number of possible actions to choose from in each state of the environment. Meaning that, for every state in the Markov Decision Process (MDP), there is a finite set of actions that the agent can take.

Policy iteration contains a three different steps:

1. **Initialization:** Start with an arbitrary policy π_0 and initialize the value function $V_0(s)$ for all states s .
2. **Policy Evaluation:** For the current policy π_k , the value function $V_k(s)$ is computed for all states s using the Bellman equation:

$$V^\pi(s) = \sum_a \pi(a|s) \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right]$$

However, for a deterministic policy, the action is chosen with probability 1, so the equation simplifies to:

$$V^\pi(s) = R(s, \pi(s)) + \gamma \sum_{s'} P(s'|s, \pi(s)) V^\pi(s')$$

3. **Policy Improvement:** The policy π_{k+1} is updated by choosing the action that maximizes the expected value for each state:

$$\pi'(s) = \arg \max_a \left[R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right]$$

After a number of iterations, the policy will converge to the optimal policy: $\pi = \pi^*$, and the value function will converge to the optimal value function $V = V^*$.

Main idea: Policy Improvement cannot make the policy worse

Define the action-value function under policy π as:

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')$$

Since V^π is the value of following π ,

$$V^\pi(s) = Q^\pi(s, \pi(s))$$

The policy improvement step chooses:

$$\pi'(s) = \arg \max_a Q^\pi(s, a)$$

Therefore,

$$Q^\pi(s, a) = R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s')$$

So we have

$$R(s, \pi'(s)) + \gamma \sum_{s'} P(s'|s, \pi'(s)) V^\pi(s') \geq V^\pi(s) \quad (1)$$

This says that taking the improved action once, and then following the old policy, is at least as good as following the old policy immediately.

From (1), we apply the reasoning recursively to get:

$$V^\pi(s) \leq \mathbb{E}_{\pi'} [R_0 + \gamma R_1 + \gamma R_2 + \dots]$$

On the right side we simply get the expected value for following the new policy π' from s , which is $V^{\pi'}(s)$. So we have,

$$V^{\pi'}(s) \geq V^\pi(s), \quad \forall s \in \mathcal{S}$$

the algorithm will eventually converge to the optimal policy because of the finite number of states and actions $\Rightarrow$ (finite number of policies) so the policy improvement must eventually converge to the optimal policy.

Basically the idea of policy improvement is based of four main points:

- Policy improvement $\Rightarrow V^{\pi_{k+1}} \geq V^{\pi_k}$
- Finite number of policies $\Rightarrow$ algorithm must terminate
- Termination $\Rightarrow$ no action can improve the policy
- No improving action $\Rightarrow$ Bellman optimality equation $\Rightarrow \pi = \pi^*$

1.2 Question 2

Q2: “Prove that the Bellman optimality equation has a unique solution for a finite MDP with a bounded reward function.”

Setup

Consider a finite MDP with state space $\mathcal{S}$, finite action sets $\mathcal{A}(s)$, transition probabilities $P(s' | s, a)$, discount factor $\gamma \in [0, 1)$, and a *bounded* reward function: there exists $R_{\max} < \infty$ with $|R(s, a, s')| \leq R_{\max}$ for all s, a, s' .

Since $\mathcal{S}$ is finite, a value function is just a vector $V \in \mathbb{R}^{|\mathcal{S}|}$. We measure distance between two value functions with the **max-norm**:

$$\|V - U\|_{\infty} = \max_{s \in \mathcal{S}} |V(s) - U(s)|.$$

This space is *complete* (finite-dimensional real vector spaces always are).

Define the **Bellman optimality operator** T , which takes a value function V and produces a new one:

$$(TV)(s) = \max_{a \in \mathcal{A}(s)} \sum_{s'} P(s' | s, a) [R(s, a, s') + \gamma V(s')].$$

A solution to the Bellman optimality equation is exactly a **fixed point** of T , i.e. a V^* with $TV^* = V^*$.

Step 1: T is a contraction

Theorem 1.1 For any two value functions V, U ,

$$\|TV - TU\|_\infty \leq \gamma \|V - U\|_\infty.$$

Proof. Fix a state s . Both $(TV)(s)$ and $(TU)(s)$ are maxima over the same finite action set, so their difference is bounded by the largest per-action gap:

$$|(TV)(s) - (TU)(s)| \leq \max_a \left| \sum_{s'} P(s' | s, a) \gamma (V(s') - U(s')) \right|.$$

(Intuitively: whichever action attains the max for V can be no more than as far “ahead” of U ’s best action as the two functions ever differ, and vice versa the reward terms are identical for V and U and cancel out.)

Since $P(\cdot | s, a)$ is a probability distribution, this average is at most γ times the worst-case disagreement between V and U :

$$\left| \sum_{s'} P(s' | s, a) \gamma (V(s') - U(s')) \right| \leq \gamma \sum_{s'} P(s' | s, a) |V(s') - U(s')| \leq \gamma \|V - U\|_\infty.$$

This bound holds for every state s , so taking the max over s gives

$$\|TV - TU\|_\infty \leq \gamma \|V - U\|_\infty. \quad \square$$

Since $\gamma < 1$, this says T strictly shrinks distances exactly like the photocopier at $\gamma \times 100\%$ zoom.

Step 2: Contractions have a unique fixed point

Theorem 1.2 – Banach Fixed-Point Theorem If T is a contraction on a complete space (i.e. $\|TV - TU\| \leq \gamma \|V - U\|$ for some $\gamma < 1$), then T has **exactly one** fixed point, and repeatedly applying T from any starting point converges to it.

We’ve shown T satisfies exactly this hypothesis. So this theorem hands us existence, uniqueness, and convergence all at once, with no extra work.

A direct one-line uniqueness check (optional, self-contained)

Suppose V_1 and V_2 are both solutions: $TV_1 = V_1$ and $TV_2 = V_2$. Then by Step 1,

$$\|V_1 - V_2\|_\infty = \|TV_1 - TV_2\|_\infty \leq \gamma \|V_1 - V_2\|_\infty.$$

Since $\gamma < 1$, the only number $d \geq 0$ satisfying $d \leq \gamma d$ is $d = 0$. So $\|V_1 - V_2\|_\infty = 0$, meaning $V_1 = V_2$.

Conclusion

Theorem 1.1 For a finite MDP with bounded rewards and $\gamma \in [0, 1)$, the Bellman optimality equation has a unique solution V^* .

Proof. T is a contraction (Step 1) on a complete space, so by the Banach Fixed-Point Theorem (Step 2) it has exactly one fixed point i.e. the Bellman optimality equation has exactly one solution. $\square$

Bonus: this same argument is why *value iteration* works. Value iteration is literally $V_{k+1} = TV_k$ repeatedly photocopying. Banach's theorem guarantees this converges to V^* no matter what V_0 you start from.

This answer is partially inspired by the explanation in [1].

1.1 Question 3

Q3: “Analyze the challenges of using linear function approximation in value-based reinforcement learning. Discuss the potential issues that may arise and propose methods to mitigate them”

2 Practical Part

Actions and rewards describing the grid-world can be found in the course book at page 60.

References

- [1] P. Salunkhe, “Understanding bellman operators: The engine of dynamic programming.” <https://medium.com/@salunkheprathmesh0/understanding-bellman-operators-the-engine-of-dynamic-programming-96eb9cb06e93>, 2023. Accessed: 2026-09-15.